23 **A** Since X and Z carry the same current of 3.0 A and are the same distance from Y, the magnitude of the forces $F_{\text{on Z due to Y}}$ and $F_{\text{on X due to Y}}$ are the same.

$$\frac{F_{\text{on Z due to Y}}}{L_z} = \frac{\mu_0 I_Y}{2\pi d} \times I_Z = \frac{\mu_0 \times 5.0}{2\pi d} \times 3.0 = 8.0 \times 10^{-6} \text{ N m}^{-1}$$

Since the I_X and I_Y are in same directions, the two wires attract each other and the force per unit length acting on X due to Y is 8.0×10^{-6} N m^{-1} to the right.

$$\frac{F_{\text{on X due to Z}}}{L_z} = \frac{\mu_0 I_Z}{2\pi(2d)} \times I_X = \frac{\mu_0 \times 3.0}{2\pi(2d)} \times 3.0 = 8.0 \times 10^{-6} \times \frac{1}{2} \times \frac{3.0}{5.0}$$

$$= 2.4 \times 10^{-6} \text{ N m}^{-1} \text{ to the left}$$

Resultant force per unit length on wire X = $(8.0 - 2.4) \times 10^{-6} = 5.6 \times 10^{-6}$ N m^{-1} to the right